MILITARY SPECIFICATION SHEET

ELECTRON TUBE, KLYSTRON

TYPE 6390

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Reflex oscillator, external cavity, pulsed operation, frequency range 6.7 to 11.065 GHz

ABSOLUTE RATINGS:

Parameter:	Ef	Ec1	Ec2	Ec3	Er	Ik	Ic1	T(g3)
Unit:	V	Vdc	Vdc	Vdc	Vdc	mAdc	mAdc	°C
Maximum:	6.8	25	1,250	1,250	-50	21	6.0	160
Minimum:	5.7	---	---	---	-700	---	---	---

PHYSICAL CHARACTERISTICS:

Dimensions: See figure 1
Base: Special 4-pin of mica-filled bakelite
(E) Mounting position: Any

TEST CONDITIONS:

Parameter:	Ef	Ec1	Ec2	Ec3	Ik	Er
Unit:	V	Vdc	Vdc	Vdc	mAdc	Vdc
	6.3	Note 2	1,250	1,250	20	Adj for max Po

GENERAL:

Qualification - Required

(E) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS MINIMUM	LIMITS MAXIMUM	UNIT
Ⓔ	Qualification inspection						
4027	Temperature coefficient	9	Ec2 = Ec3 = 1,000 Vdc	ΔF °C	---	±0.1	MHz °C
	Quality conformance inspection, part 1	1					
1256	Electrode current (control grid)	2		Ic1	---	6	mAdc
1261	Electrode voltage (control grid)	2		Ec1	2	16	Vdc
4214	Cathode emission	7	Cavity VII: Ef = 5.8 V	ΔIk/Ik	---	10	%
4229	Total reflector current	-	Nonoscillating: Er = 300 ± 100 Vdc	Ir	---	7	μAdc
4250	Power output	3, 5	Cavity VA; 3-3/4 mode: Er = -285 to -420 Vdc	Po	45	---	mW
		3, 5	Cavity VA; 4-3/4 mode: Er = -115 to -200 Vdc	Po	45	---	mW
		4, 5, 6	Cavity VII; 4-3/4 mode: Er = -285 to -420 Vdc	Po	45	---	mW
4213	Reflector voltage	3, 5	Cavity VA; 3-3/4 mode	Er	-420	-285	Vdc
		3, 5	Cavity VA; 4-3/4 mode	Er	-220	-115	Vdc
		4, 5, 6	Cavity VII; 4-3/4 mode	Er	-360	-220	Vdc
---	Frequency	3, 5	Cavity VA; 3-3/4 mode: Er = -285 to -420 Vdc	F	8,950	9,450	MHz
		3, 5	Cavity VA; 4-3/4 mode: Er = -115 to -200 Vdc	F	8,950	9,450	MHz
		4, 5, 6	Cavity VII; 4-3/4 mode: Er = -285 to -420 Vdc	F	10,515	11,065	MHz
	Quality conformance inspection, part 2						
1031	High-frequency vibration	8	No voltages	---	---	---	---
1261	Electrode voltage (control grid at Ib cutoff)	-	Ef = 6.3 V; Ec2 = Ec3 = 1,250 Vdc; Ib = 1 mAdc (max)	Ec1	---	-23	Vdc
Ⓔ 4289	Heater current (non operating)	-	Ef = 6.3 V	If	520	640	mA
---	Pulsed operation	10	3-3/4 mode	F	6,870	9,200	MHz
		10	4-3/4 mode	F	9,150	10,790	MHz
4213	Reflector voltage	4, 5, 6	Cavity III; 3-3/4 mode	Er	-195	-90	Vdc
4250	Power output	4, 5, 6	Cavity III; 3-3/4 mode	Po	45	---	mW
---	Frequency	4, 5, 6	Cavity III; 3-3/4 mode	F	6,700	7,040	MHz
	Quality conformance inspection, part 3						
---	Life test	-	Group D	t	500	---	hrs
---	Life-test end point:						
4250	Power output	3, 5	Cavity VA; 3-3/4 mode; Er = -285 to -420 Vdc;	Po	36	---	mW
		3, 5	Cavity Va; 4-3/4 mode; Er = -115 to -200 Vdc;	Po	36	---	mW
		4, 5, 6	Cavity VII; 4-3/4 mode; Er = -285 to -420 Vdc;	Po	36	---	mW

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NOTES:

1. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.
2. Ec1 shall be adjusted for a total current (Ib) of 20 mA to the interaction gap electrodes with the TUT in oscillation. The reflector voltage (Er) shall be adjusted for maximum power output.
3. The TUT shall be mounted in the specified test cavity as shown on Drawing 177-JAN.
4. The TUT shall be mounted in the specified test cavity as shown on Drawing 176-JAN.
5. The reflector voltage (Er) shall be adjusted between the specified limits in the specified reflector transit mode for maximum power output into a fixed 50-ohm load with VSWR < 1.25.
6. With the TUT mounted in the specified cavity, the loop shall be adjusted for maximum power output into a fixed 50-ohm load with VSWR < 1.25.
7. The TUT shall be oscillating for this test.
8. After performance of the specified test, the TUT shall satisfy the requirements of quality conformance inspection, part 1, as specified herein.
9. With the test cavity shown on Drawing 177-JAN, the temperature coefficient of the cavity shall be -11 kHz/°C maximum. The temperature coefficient of the TUT shall be considered as being the difference between the characteristics of the cavity alone and TUT with the cavity.
10. The TUT shall be oscillating in the specified transit mode during this test.
    (a) The jitter or multiple shift in the rise time shall be less than 0.2 μs at any frequency in the specified frequency range, and there shall be no evidence of loss of pulse or intermittent pulsing. The top of the pulse wave shape as viewed on a suitable oscilloscope shall be essentially flat.
    (b) The pulse amplitude shall not vary more than 1 ± 0.2 dB of the CW amplitude.
    (c) Pulse shortening in each mode over the specified frequency range shall not exceed 0.4 μs of the output viewed pulse.
    (d) Jitter pulse shortening shall not exceed 0.4 μs throughout each mode specified herein.
Ⓔ (e) The pulse repetition rate shall be varied between 40 and 4,000 pulses per second with a pulse duration of 0.5 μs.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MU
Air Force - 80
DSA - ES

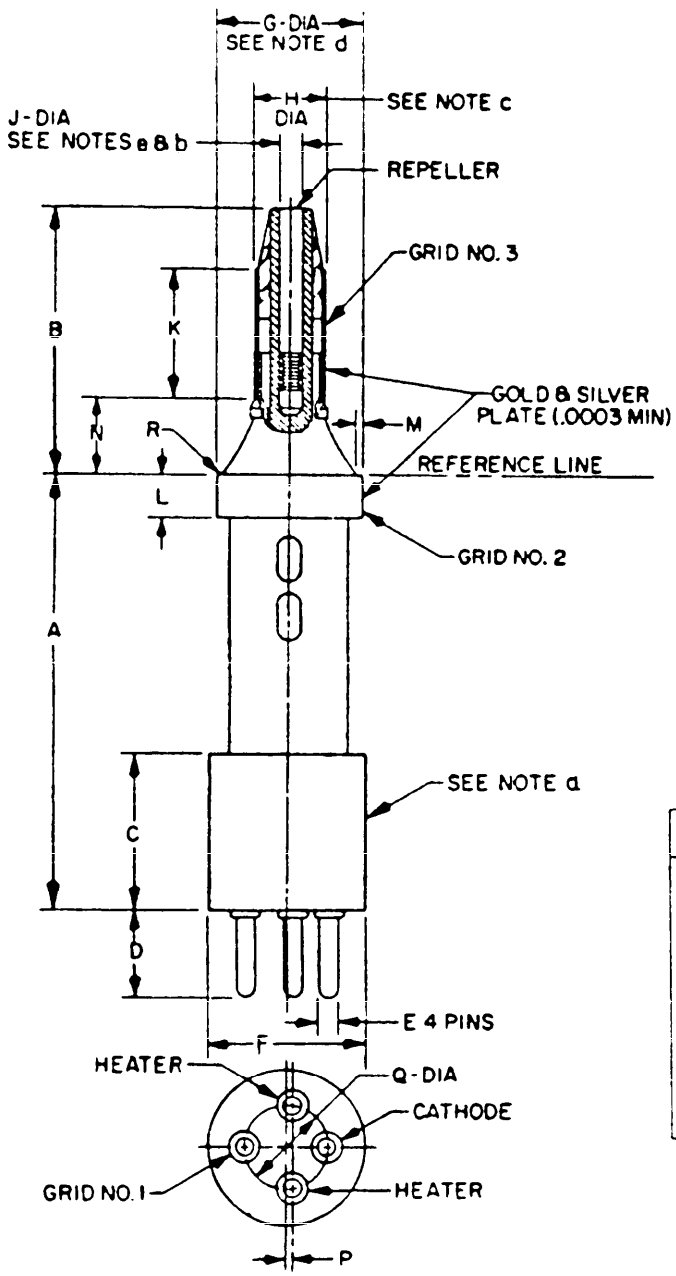
User activities:
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Navy - AS, OS, MC, CG
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-3064)

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Electrical connections	
Pin No.	Element
1	g
2	h
3	k
4	h
Lower ring	g_2
Upper ring	g_3
Jack	r

FIGURE 1. Outline drawing of electron tube type 6390.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Qualification inspection		
C	.720(18.29)	.780(19.81)
D	.467(11.86) Nom	
E	.090 (2.29)	.097 (2.46)
F	.782(19.86)	.844(21.44)
P	.030 (.76) Nom	
Q	.415(10.54) Nom	
Quality conformance inspection, part 1 (see note 1)		
H	.372 (9.45)	.376 (9.55)
M	.020 (.51)	
N	.340 (8.64)	.407(10.34)
R	.003 (.08)	.008 (.20)
Quality conformance inspection, part 2		
A	1.860(47.24)	2.360(59.94)
B	1.360(34.54)	1.420(36.07)
G	.782(19.86)	.786(19.96)
J	.105 (2.67)	.113 (2.87)
K	.575(14.61)	.675(17.15)
L	.190 (4.83)	.230 (5.84)

NOTES:
a. Low-loss phenolic base.
b. This diameter applies for a depth of .312(7.92 mm) min.
c. Reference diameter.
d. This diameter to be concentric to H within .005(.13 mm).
e. This diameter to be concentric to H within .015(.38 mm).

FIGURE 1. Outline drawing of electron tube type 6390 - Continued.

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